



Pramana

BRSR · E1 INTELLIGENCE

Every answer on India's climate disclosures, *with its proof.*

Pramana is a small language system built from scratch for the IIMB BRSR study. It answers questions on the GHG emissions and climate-risk disclosures of 982 listed companies, and every figure it shows links to the exact workbook cell, rating, report table or report page it came from. The same question always returns the same answer, and anything outside the data is refused rather than guessed.

982 companies, 22 NSE sectors	38 E1 BRSR questions covered	597 / 597 report figures reproduced exactly	0 uncited numbers across 178 tests
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**Pramana**
BRSR · E1 INTELLIGENCE

+ New question

YOUR COMPANY LENS
• Not set, answers are general

Ask

Companies

Sectors

Method and sources

CONVERSATIONS
0
Your questions are remembered in this browser only.

New question

Find a company

Set lens

BRSR FY 2024-25 · THEME E1 · GHG EMISSIONS AND CLIMATE RISK

Ask the record.

Get the *proof.*

Pramana answers questions on the climate disclosures of India's top listed companies using only the IIMB dataset. Every figure links to the exact workbook cell, rating or report page it came from. If it is not in the data, it says so.

982
companies with BRSR filings

22
NSE sectors benchmarked

1.31B
tCO₂e Scope 1+2 disclosed

597/597
report figures reproduced

START WITH A QUESTION

or press ⌘K to find a company

PLAIN DATA
What are Tata Steel's Scope 1 and Scope 2 emissions?
Ask →

PEER COMPARISON
How does ACC compare with its peers?
Ask →

BEST PRACTICES
Best practices for GHG reduction projects in cement
Ask →

Ask about any of the 982 companies, sectors or the E1 report

LIVE APP

pramana-brsr.onrender.com

CODE

github.com/RS-010806/BRSR_SLM

DATA

E1 data.xlsx (Questions, Base Data, Rating), E1 chapter, IIMB report *Business Responsibility and Sustainability in India*, FY 2024-25

PREPARED FOR

TCI-IIMB Supply Chain Sustainability Lab, Supply Chain Management Centre, IIM Bangalore · 30 September 2026

What was asked, and how Pramana answers it

The meeting set five hard requirements: answer only from the provided data, never hallucinate, be deterministic, show visuals rather than plain text, and run as a cloud web app. Each maps to a concrete design decision.

REQUIREMENT	HOW IT IS MET
Plain data such as Scope 1 and 2 for any company	All 38 E1 questions for all 982 companies, with the value as filed, the prior year, the change, the Rating-sheet score and where it sits on the rubric. Delivered
Peer comparison by industry sector	Seven-dimension peer positioning with percentiles, full sector rankings, side-by-side scorecards for up to six companies, cross-sector views. Delivered
Best practices from qualitative disclosures	Top-rated disclosures quoted verbatim with their most specific sentences highlighted, plus a count of the practices that distinguish leaders from the rest. Delivered
One or two non-hallucinated examples in best-practice answers	Three examples per answer, selected deterministically, each reproduced character for character from the source cell with its address (for example Base Data AE817). Delivered
Dashboards and graphs , not just text	Twenty-two block types: KPI tiles, ranked bars, distribution strips, peer-position rows, score grids, treemaps, rubric scales, verbatim quote cards, a live what-if simulator. Every chart has a table view and CSV export. Delivered
Strict guardrails : say plainly when something is not there	No web access and no free-text generation. Other ESG themes, finance, forecasts, advice and prompt injection are refused with a pointer to the right chapter of the report; absent companies are named, never substituted. Delivered
Deterministic : the same prompt never gives conflicting answers	No sampling anywhere. Identical question and context produce byte-identical output, stamped with an answer fingerprint anyone can compare. Verified in tests across fresh engine instances. Delivered
Chat history and follow-ups , no user information collected	Conversations persist in the browser; follow-ups such as "and Scope 3?" reuse context. An optional company lens personalises "we" and "our" without sign-in. Delivered
Cloud web app , not a local install	Deployed on Render from a Docker blueprint; about 2 ms per answer and 160 MB of memory, so it runs on the free plan. Delivered
Initial scope of the environmental questions	All E1 questions in the workbook (35 with company data, plus rating-only and green-credit rows, which are reported as unpopulated). Built to extend to the full question bank. Delivered

Open question from the meeting, *"can the system reproduce verbatim examples of best practices without hallucinating?"*: yes. Examples are not generated at all; they are selected spans of the filed text, and the evidence drawer shows the full cell so the context can be checked.

A language system that understands freely and answers strictly

A generative model cannot guarantee zero hallucination or identical answers.

Pramana therefore uses a small neural model only to understand the question, and answers purely by computation over the data.

01

Normalise

Unicode, possessives, "Scope 1/2/3", fiscal years and numbers are made canonical, identically in training and use.

02

Link

A token trie over 2,664 company aliases, 22 sectors, 38 questions and out-of-scope topics, with typo tolerance.

03

Understand

A transformer trained from scratch predicts one of 16 intents and a topic. Explicit rules override it where text is unambiguous.

04

Compute

Handlers compute values, ranks, medians, distributions and scenarios using the report's own conventions.

05

Prove

Answers are assembled from templates; every numeric statement must carry a citation before it is returned.

The small language model

Tokenizer	Byte-pair encoding written from scratch, 1,472 symbols
Architecture	Pre-LN transformer encoder, 2 layers, 4 heads, d = 128, intent and topic heads
Parameters	464,299
Training data	52,250 queries generated from a hand-written grammar of 272 templates, with typo and word-drop noise
Accuracy	99.3% intent on synthetic validation; 110/110 on hand-written queries for intent, company, sector and metric
Inference	Pure numpy, CPU only, matches the PyTorch model to 2.4e-7

Why this is safe for high-stakes use

- The model outputs a label, never text, so it has nothing to invent.
- Company, sector and metric names are resolved against the dataset; unknown or absent names stop the answer.
- Numbers are calculated from the workbook at answer time and formatted by fixed rules.
- A grounding check rejects any paragraph with a number but no citation; the test suite enforces it on every query.
- The answer fingerprint (SHA-256 of content plus dataset version) proves two answers are identical.
- The server is stateless: conversation context travels with each request and nothing about the user is stored.

Answer ID 0d3bd07dfbb963a0 • dataset 7dfc85ed9606

How I understood this

HOW THIS WAS UNDERSTOOD AND ANSWERED

1 NORMALISE

how does <co>
compare with its
peers

2 CLASSIFY

peer_benchmark 96.4%
compare 0.3%
text_search 0.3%

3 LINK

Company: Tata Steel Limited

4 GROUND

1 citation
2 numeric statements, 0 uncited
3.1 ms + 0.8 ms

Final route: peer_benchmark • model topic none (97%)

BPE pieces: how</w> does</w> <co> compare</w> with</w> its</w> peers</w>

"How I understood this" opens the trace for any answer: the normalised tokens with entity placeholders, the model's top three intents with probabilities, the linked entities, the rules that fired, the grounding check and timing.

597 of 597 published figures reproduced from raw data

A reconciliation module recomputes every table in the E1 chapter from the workbook and compares it cell by cell. Matching them exactly required recovering the report's unstated conventions, which Pramana now applies everywhere.

Conventions recovered

- Yes/No counts come from the Rating sheet (100 = Yes), which is what the report used.
- Sector totals exclude SIS Limited and Patel Engineering's previous-year Scope 2, both implausible billion-tonne values.
- Company-wise direction counts an unchanged total as "increased" (Table 3.3).
- Change bands are < -10, -10 to -5, -5 to +5, +5 to +10 and > +10 percent, boundaries as published.
- Per-rupee intensity is shown per crore, as in Table 3.5.
- The year-on-year rating rubric reproduces 100% of rated cases.

Issues found in the source data

- Base Data columns for Q241 and Q268 carry headers from other principles; answers use the Rating sheet.
- Hindustan Unilever is classified as Metals & Mining and Hindustan Zinc as Consumer Durables: adjacent rows that appear swapped.
- 14 companies, including Tata Steel (61 tCO₂e Scope 1), filed values below 0.1% of their sector median, likely in thousand or million tonnes.
- 213 companies report per-rupee intensity implying over 10,000 tCO₂e per crore.
- The Rating sheet gives 100 when both years are 0, and its AI-scored questions differ from the report's tables (a later scoring run).

Principle kept: the report's "as-reported" fidelity. Nothing is corrected or imputed. Suspect values are shown exactly as filed, flagged with the specific numbers behind the flag, and kept out of level rankings where they would mislead. Year-on-year changes, which are unit-invariant, are still compared.

• TATA STEEL LIMITED • METALS & MINING

Scope 1 emissions

⚠ DATA QUALITY Magnitude check: reported Scope 1+2 of 66.00 tCO₂e is below 0.1% of the Metals & Mining median (187,684 tCO₂e), which suggests it was filed in a scaled unit such as thousand or million tCO₂e. It is shown exactly as filed and counts in sector totals as in the report, but it is kept out of level rankings and peer percentiles. Year-on-year change is unaffected.

Tata Steel Limited reported Scope 1 emissions of 61 tCO₂e in FY 2024-25 ⁽¹⁾, against 59 tCO₂e in FY 2023-24 ⁽²⁾, a change of +3.4%.

The Rating sheet scores this year-on-year movement 25/100 (rose by up to 5% year on year) ⁽³⁾.

Data-quality flag: the filed Scope 1+2 total is below 0.1% of the Metals & Mining median, which suggests a scaled unit (thousand or million tCO₂e). The value is shown exactly as filed and is not ranked against peers ⁽⁴⁾.

Its year-on-year change of +3.4% compares with a Metals & Mining median change of -2.2% across 20 companies ⁽⁵⁾.

FY 2024-25	FY 2023-24	Year on year	Rating Q1330
61 tCO ₂ e	59 tCO ₂ e	+3.4% change vs FY 2023-24 rose	25 out of 100

How Q1330 is scored
Scope 1 emissions, FY 2024-25

0 rose by more than 5% year on year 6 companies

Ask about any of the 982 companies, sectors or the E1 report

No sign-in. Memory stays in this browser. Enter to send, Shift+Enter for a new line

Answers come only from the IIMB E1 dataset and report. Figures are as filed by companies for FY 2024-25.

Tata Steel's filed Scope 1 of 61 tCO₂e is shown as filed, with a magnitude-check flag instead of a misleading "lowest emitter" ranking.

• OUTSIDE THIS DATASET

Water management is not part of E1

Not available here. Water management for Tata Steel is covered by chapter E3 of the IIMB report ⁽¹⁾, but this assistant is loaded only with the E1 data on GHG emissions and climate risk, so I cannot give company-level figures for it.

For Tata Steel, I can show Scope 1, Scope 2 and Scope 3 emissions, intensity, assurance, targets and GHG reduction projects.

For context, the IIMB report's executive summary on E3 (Water Management) states

Water demand is expanding, not contracting: 33.60% of companies show >+10% withdrawal increase year-on-year, with median withdrawal growth of +4.64% and consumption growth of +3.65%. 76.07% of companies consume 80–100% of all water withdrawn, with an all-sector median of 100% — confirming that corporate water use is almost entirely consumptive with minimal return flows.

Executive summary, E3 PDF page 8 of 2 ⁽²⁾

Show Tata Steel's E1 profile What are Tata Steel's Scope 1 emissions? How does Tata Steel compare with its peers?

2 sources every figure above links to one of these

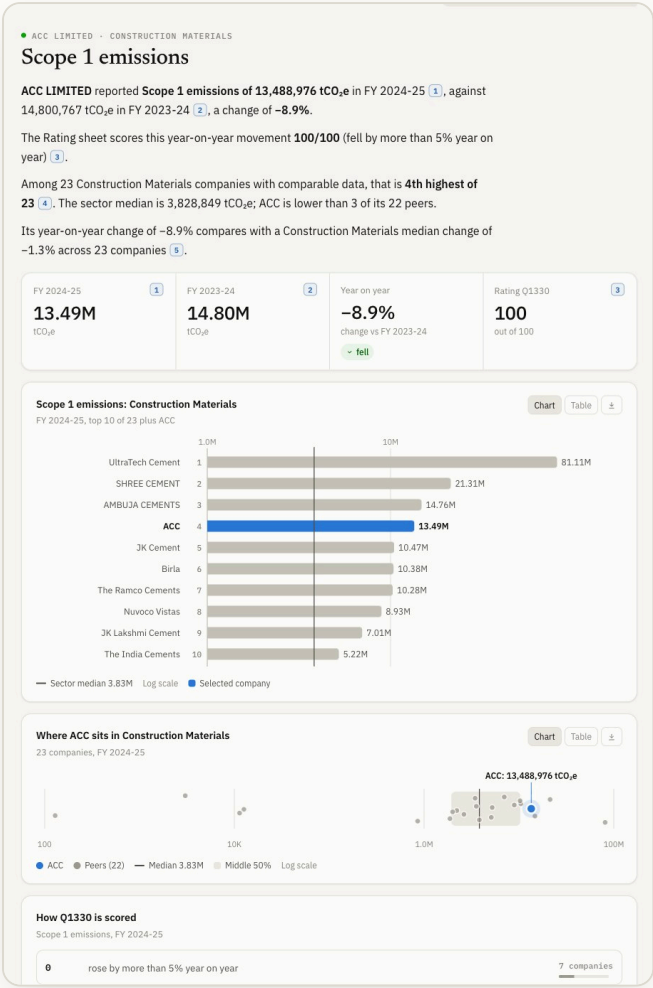
Answer ID df6619829b48fed3 dataset 7dfc85ed9606

How I understood this

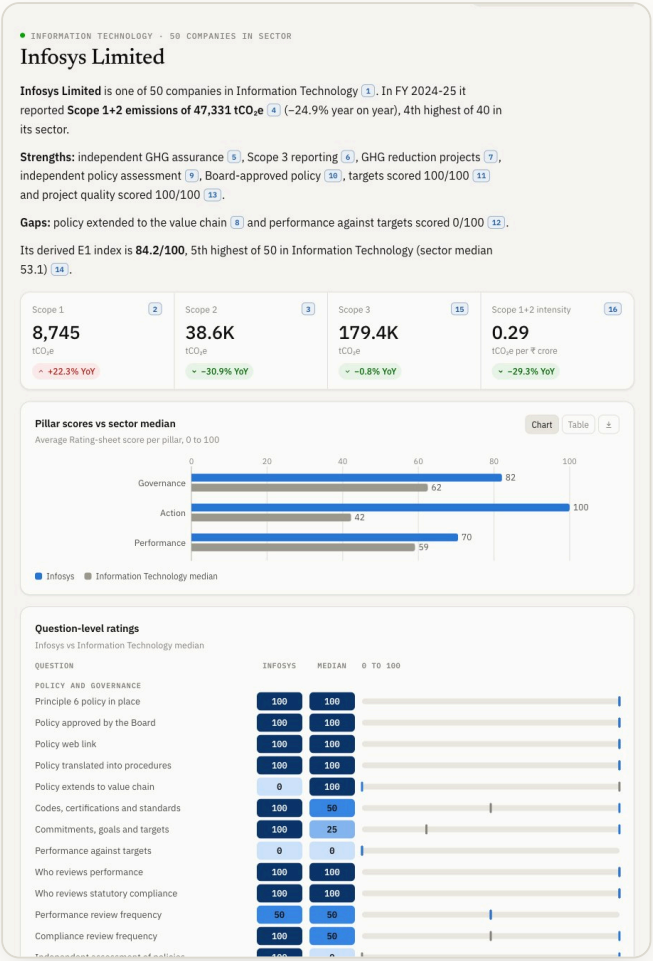
Ask about any of the 982 companies, sectors or the E1 report

Questions outside E1 are refused plainly, with the chapter that covers them and the report's own executive-summary finding, quoted verbatim.

Plain data, with its rating and its peers



A single metric. Value as filed, prior year and change, the Rating-sheet score in plain words, the sector rank and median, the top ten with the median marked, the full distribution and the rubric with peer counts at each level.

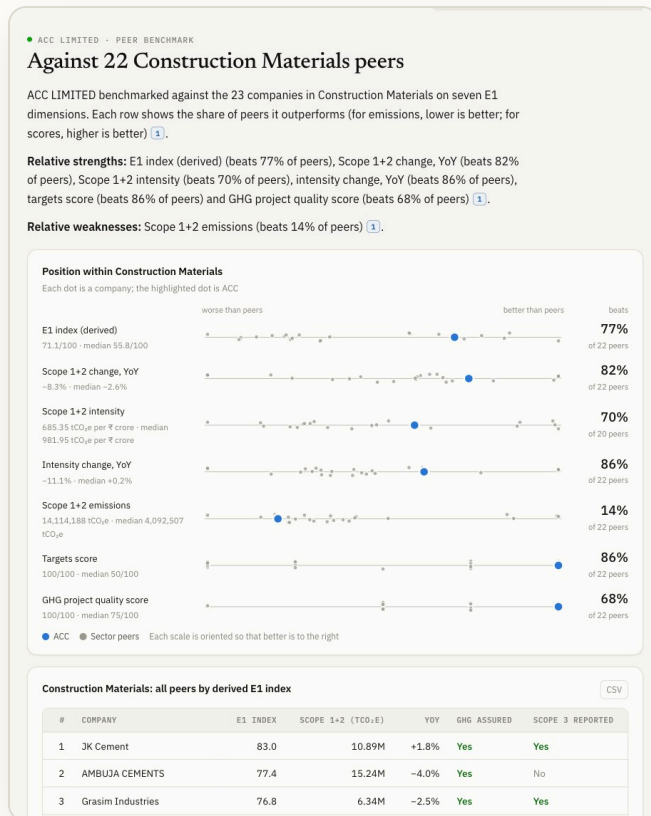


Company profile. Headline emissions and rank, strengths and gaps with each claim cited, KPI tiles, pillar scores against the sector median, and a rating grid labelled by what each score measures (change, or level against the industry).

Sixteen kinds of question, each with its own answer layout

Company profile "Tell me about Hindalco"	Single metric "Maruti Scope 3 last year"	Compare "TCS vs Infosys vs Wipro"	Peer benchmark "Cipla vs other pharma"
Sector overview "Banking sector snapshot"	Ranking "Greenest IT companies"	Best practice "Good examples of net zero targets"	Aggregate "How many companies report Scope 3?"
Screen "Cement companies without assurance"	Disclosure search "Who mentions SBTi?"	What-if "If ACC cut intensity 20%?"	Report insights "Summarise the E1 chapter"
Explain "How is Scope 1 rated?"	Company lens "I work at ACC"	Refusal "Tata Steel's revenue?"	Help "What can you do?"

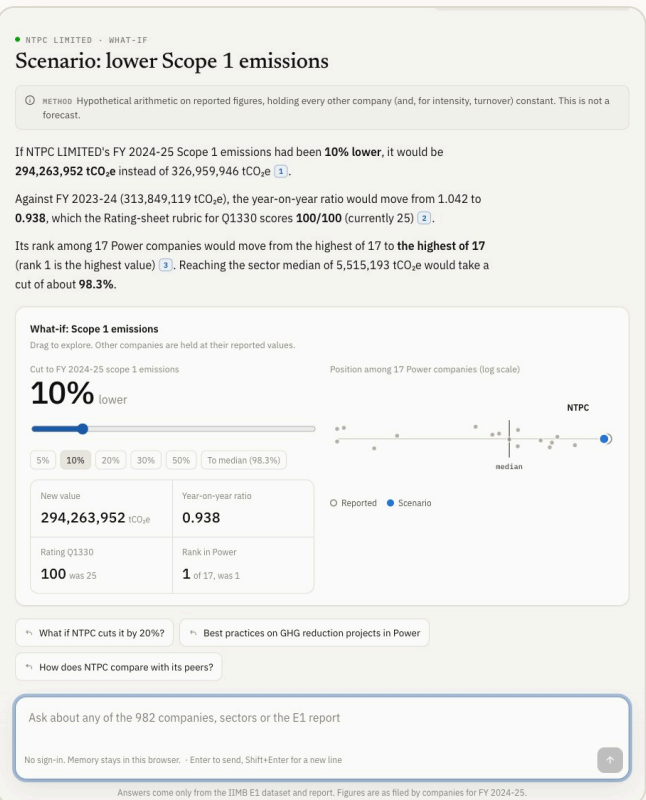
Peer benchmarking and what-if scenarios



Peer benchmark. Seven dimensions, each oriented so better is to the right, with the share of sector peers the company outperforms and the full peer table.

Personalised, without sign-in

Say "I work at ACC" or set the company lens. Questions like "how do we compare with our peers?" and "what can we learn from the leaders?" then use that company, and best-practice answers add a gap view: the company's score and specificity against the leaders, and the practices common among top-rated peers that its own disclosure does not mention.



What-if simulator. Drag a reduction and see the new value, the Rating-sheet score it would earn, the sector rank and the cut needed to reach the median. Labelled as arithmetic on reported figures, not a forecast.

Honest about comparability

Absolute emissions reflect business models, so cross-sector comparisons add a context note and point to intensity. Physical-output intensity uses company-specific units, so only its change is compared. Scaled-unit filings are compared on change but never on level.

Best practices, reproduced verbatim

• BEST PRACTICES • CONSTRUCTION MATERIALS

GHG reduction project details

In Construction Materials, **8 of 23 companies** score 100 on GHG reduction project details, the level described as “Projects are highly impactful, innovative, and large in scale; achieve significant, measurable, and time-bound GHG reductions; aligned with global best practices or standards (e.g., SBTi, Net Zero pathways)” (1). The most specific of these disclosures are reproduced verbatim below, with their most concrete sentences highlighted.

Examples: **Star Cement** (Construction Materials) (2), **UltraTech Cement** (Construction Materials) (4) and **JK Cement** (Construction Materials) (6).

Practices named far more often by top-rated companies than by the rest: low-carbon products and clinker substitution (100% vs 36%), energy efficiency (88% vs 43%), waste heat recovery (88% vs 50%) and science-based or net-zero targets (62% vs 0%) (5).

Top-rated disclosures, verbatim

Verbatim from the filing. Highlights mark the most specific sentences.

Star Cement 100/100
Construction Materials - Q1342 GHG reduction project details

Commissioning of 12 MW Waste Heat Recovery System (WHRS) thereby utilizing waste gases from the kiln and clinker cooler to generate electricity. This initiative significantly reduces dependency on fossil fuels-based power sources, reduction power cost and reduction in carbon footprint.

Installation of Variable Frequency Drives (VFDs) across multiple process areas to optimize fan operations and reduce energy demand:

Mill Reject Bag Filter Fan (561FN1): 30 KW VFD – savings 10KW/hr. (running at 30Hz)

Mill Feed Circuits (531FN1 & 531FN2): 2X 37 KW VFDs – saving 16 KW/hr

Fly Ash Circuit (K21FNE & K21FNA): 30 KW & 37 KW VFDs – saving 13 KW/hr.

Clinker Unloading Bag Filters (491FNS & 491FN1): 30 KW & 55 KW VFDs – savings 28 KW/hr. (This saving is against to

Energy efficiency Waste heat recovery Low-carbon products and clinker substitution

Base Data - AEB17 (2) (3) Show full text

UltraTech Cement 100/100
Construction Materials - Q1342 GHG reduction project details

The Company has targets to reduce 27% of its Scope 1 emission intensity and 69% of its Scope 2 emission intensity by 2032 from base year 2017, validated by Science Based Targets initiative (SBTi). Under the GHG emission reduction initiatives, Company has implemented various projects in areas of energy efficiency, circular economy, low carbon products development, renewable energy and waste heat recovery and is exploring other technological advancements including kiln electrification, electrolysis of limestone (Zero Carbon Lime (ZeroCAL)), carbon capture and storage.

In addition to Coolbrook, we have signed an agreement with University of California, Los Angeles (UCLA) to pilot ZeroCAL. The ZeroCAL process can eliminate nearly 98% of carbon dioxide emissions associated with limestone decomposition in cement manufacturing. UltraTech will be the first company globally to implement the ZeroCAL.

Renewable energy Energy efficiency Waste heat recovery Electrification and EVs Carbon capture

Science-based or net-zero targets Low-carbon products and clinker substitution

Base Data - AE921 (4) (5) Show full text

JK Cement 100/100
Construction Materials - Q1342 GHG reduction project details

Yes, JK Cement has implemented several projects and has a strong commitment towards reducing Green House Gas (GHG) emissions across its operations. There are four major strategies to reduce the carbon footprint:

1. Integrations of advanced technologies and practices to enhance energy efficiency.
2. Transitioning from fossil fuels to cleaner alternatives such as AFR
3. Increasing the use of renewable energy and waste heat recovery systems (WHRS) to reduce reliance on fossil power.
4. Lowering the proportion of clinker in cement by producing more blended cement.

JK Cement's near-term science-based GHG emission reduction targets have been validated by SBTi, aligning with a 1.5-degree trajectory. The company commits to reducing Scope 1 GHG emission intensity by 20.4% and Scope 2 GHG intensity by

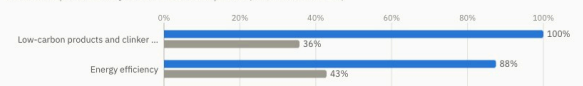
Renewable energy Energy efficiency Waste heat recovery Fuel switching and alternative fuels Carbon capture

Science-based or net-zero targets Low-carbon products and clinker substitution

Base Data - AE460 (6) (7) Show full text

Practices named in top-rated vs other disclosures

Share of companies whose Q1342 text names each practice (Construction Materials)



Ask about any of the 982 companies, sectors or the E1 report

No sign-in. Memory stays in this browser. - Enter to send, Shift+Enter for a new line

How examples are chosen

1. Take the companies scoring 100 on the question in the Rating sheet, within the sector if one is named.
2. Rank them by the specificity of their three most concrete sentences: quantities with units, years, baselines and named practices.
3. Break ties by the derived E1 index, then alphabetically. The same question always returns the same examples.
4. Quote each disclosure exactly as filed, highlight its most specific sentences, and link the cell.

What distinguishes the leaders

Practice families (renewable energy, waste heat recovery, fuel switching, science-based targets, carbon capture and others) are matched literally in the text of top-rated and other disclosures. In cement, leaders name waste heat recovery 88% of the time against 50% for the rest, and science-based targets 62% against 0%.

1 WORKBOOK CELL

DR. REDDY'S LABORATORIES LIMITED

Q286 - Commitments, goals and targets

Specific Commitments Goals And Targets Set By The Entity With Defined Timelines Explanatory Text BlockD_Principle6

We strengthened our commitment to sustainability and announced new environmental, social, and governance (ESG) goals for 2030 to make more meaningful impact through our sustainable development strategy.

A. Being committed to environmental stewardship: Reducing carbon emissions

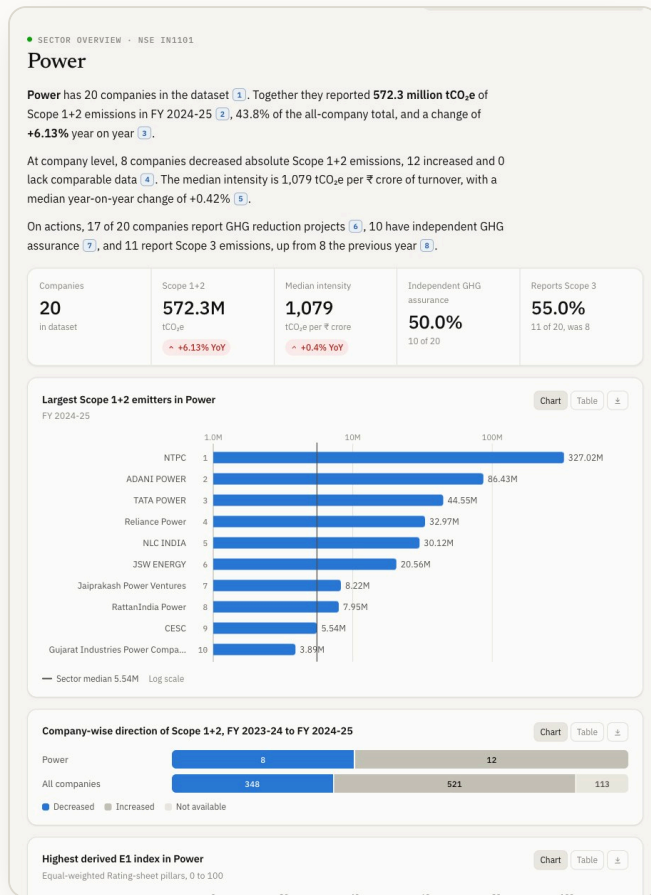
- 100% renewable power (RE100) by 2030; • Carbon neutral in direct operations (Scope 1 and 2 emissions) by 2030; • 12.5% reduction in our indirect carbon emissions (Scope 3) by 2030
- Water positivity
- Water positive by 2025

B. Making our products accessible and affordable for patients

Access: • Serve 1.5 billion+ patients by 2030

Full context, one click away. The evidence drawer shows the complete disclosure, so a highlighted sentence can always be read in context.

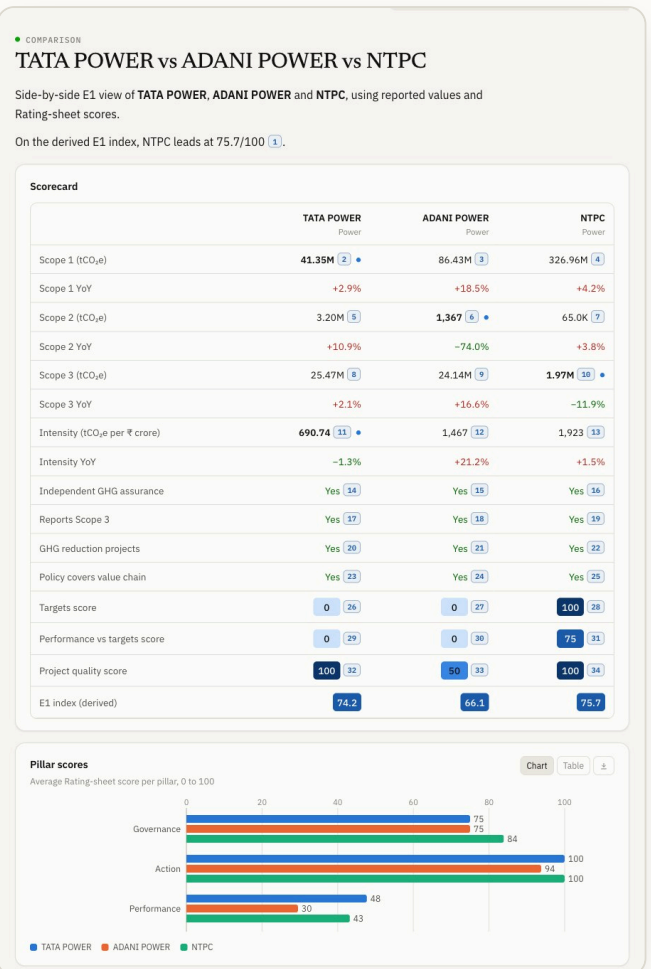
Sectors, the whole market and head-to-head comparisons



Sector overview. Totals, change, company-wise direction, intensity, assurance and Scope 3 coverage, each cited to its report table row, plus the report's own observations on that sector.

Screens, rankings and aggregates

"Which power companies don't have independent assurance?", "top 10 emitters", "who reduced emissions the most in steel?", "how many companies report Scope 3?" return ranked lists and tables with CSV export. Aggregates quote the report table row and page; rankings state how many companies were comparable and why others were excluded.



Comparison. A cited scorecard across emissions, change, intensity, assurance, targets and project quality, with the best value marked, and pillar scores.

A conversation, not a query box

YOU
Compare ACC and Ambuja on carbon intensity

YOU
and on Scope 3?

YOU
what about UltraTech?

The second question reuses both companies with a new metric; the third adds UltraTech to the same comparison. The trace shows which parts came from memory, and the same thread always resolves the same way.

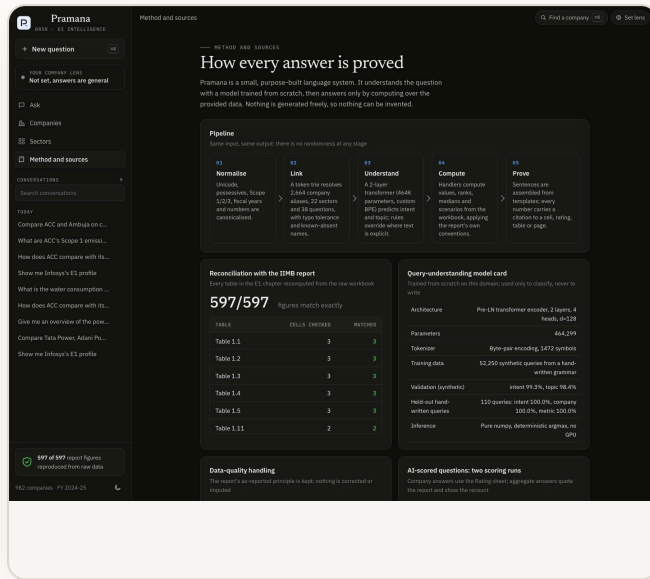
Clicking a citation opens the source. For a report table it shows the cited row with its published values and opens the report PDF at the right page. For a workbook value it shows the file, sheet and cell; for a rating, the rubric with the company's level; for a computation, the formula and its inputs, each clickable.

Disclosure search. "Which companies mention green hydrogen?" returns every company, the matching sentences and a sector split.

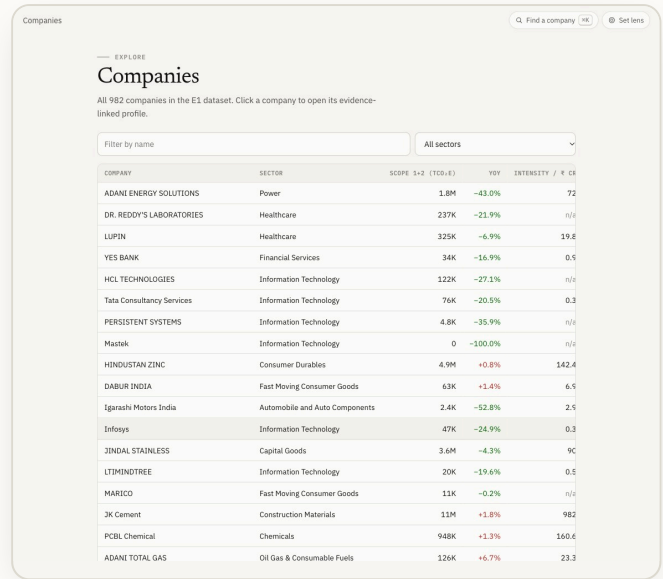
Cross-sector view. Share of disclosed emissions (Power 43.8%) as a treemap, then intensity and change by sector.

Report insights. The chapter's ten signals and conclusion, each linked to its page.

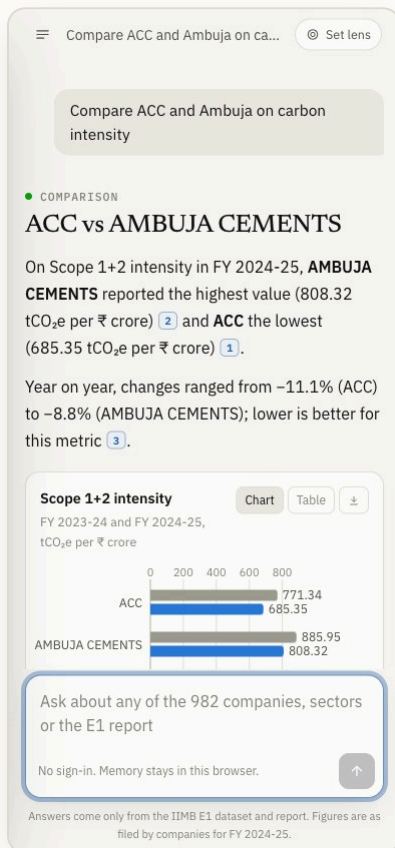
Built to be used every day



Method and sources (dark theme): pipeline, model card, reconciliation by table, data-quality handling and guardrails.



Companies directory: all 982 companies, sortable by emissions, change, intensity, assurance and index, with flags marked.



Memory and follow-ups

Conversations are saved in the browser and listed by day, searchable, deletable. Follow-ups reuse context: "and Scope 3?", "what about Ambuja?", "same for cement".

Share and export

Share links reproduce the exact answer. Copy an answer with its numbered sources, print or save it as PDF, download any chart or table as CSV.

Fast navigation

Cmd/Ctrl+K palette for any company or sector with quick actions (profile, peers, set lens). Click any bar, dot or company to drill in.

Accessible design

Colour-blind-validated palette, identity never by colour alone, table view for every chart, keyboard support, reduced-motion support, light and dark themes, phone layout.

Clarifies instead of guessing

"Adani" matches five companies, so Pramana asks which one. "ONGC" is not among the 982, and it says so rather than substituting a similar name.

Private by default

No account, name or email. The optional company lens and history stay on the device; the server stores nothing about users.

Questions to try, and what comes next

PLAIN DATA

What are Tata Steel's Scope 1 and Scope 2 emissions?

QUALITATIVE

What targets has Dr. Reddy's set?

PEERS

Where does ACC rank in its sector on emission intensity?

COMPARE

Compare TCS, Infosys and Wipro

BEST PRACTICE

Show me good examples of net zero targets

PERSONAL

I work at UltraTech · What can we learn from the leaders?

WHAT-IF

What if NTPC cuts Scope 1 by 10%?

SECTOR

Give me an overview of the cement sector

SCREEN

Which power companies don't have independent assurance?

AGGREGATE

How many companies report Scope 3 emissions?

SEARCH

Which companies mention waste heat recovery?

METHOD

How is the Scope 1 rating calculated?

Engineering

Backend	Python, FastAPI, numpy, RapidFuzz (five runtime dependencies)
Frontend	React, TypeScript, hand-built SVG charts with d3-scale
Pipeline	openpyxl, python-docx, pdfplumber; PyTorch for training only
Quality	178 tests: reconciliation, determinism, grounding, guardrails, routing, style
Speed	About 2 ms per answer; 0.4 s cold start; 160 MB memory
Deploy	Docker blueprint on Render, auto-deploys on push

Extending to the full question bank

1. Add the theme's sheets to **data/raw** and its questions to the registry.
2. Add metric phrases to the lexicon and a few templates to the grammar.
3. Rebuild and retrain (about three minutes on a laptop) and extend the reconciliation to the new chapter's tables.

Out-of-scope pointers for the other 20 parameters are already wired to their chapters, so they become answerable one theme at a time.

Limitations

- E1 only for now; other themes are refused with a pointer.
- Phrasings far from the grammar fall back to clarification rather than a guess.
- The E1 index is a derived navigation aid, not an official IIMB score.
- The free Render plan sleeps when idle; the first request after a pause is slower.